

**The Anonymity Gap: How Changing Threat Models Reshape Trust, Risk, and Behavior on
Tor**

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On my honor as a University Student, I have neither given nor received unauthorized aid on this
assignment as defined by the Honor Guidelines for Thesis-Related Assignments

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Introduction

Tor matters not merely because it is a privacy tool, but because it is used in situations where anonymity can have a material impact on safety and privacy freedoms. In journalism, anonymity-preserving systems have become part of source-protection workflows designed to reduce metadata exposure as well as content exposure. People living under surveillance likewise adapt what they search for, communicate, and organize around because they assume their activities may be observed. For survivors of intimate partner violence, failures in digital privacy can enable coercion, stalking, and physical danger. At a broader scale, Jardine (2018) argues that Tor use is especially strong in both highly repressive and highly liberal settings, suggesting that anonymity tools matter both where surveillance is feared and where privacy is actively valued. Anonymity, in this sense, is not simply a technical property. It is part of how individuals manage speech, exposure, and safety in practice.

At the same time, Tor has never guaranteed perfect invisibility. The Tor Project explicitly states that timing analysis, traffic correlation, and related deanonymization risks remain important limitations, especially under strong adversary models. EFF's user guidance similarly emphasizes that Tor Browser primarily protects activity conducted inside the browser and that ordinary user behavior can still reveal identity. What has changed is that machine-learning-based traffic analysis has made these conditions more contested. Technical work on website fingerprinting and flow correlation reports strong results in controlled settings, while later realism-oriented studies show that those results do not transfer cleanly into real-world conditions. The sociotechnical problem, therefore, is not that Tor has suddenly become useless. Rather, the meaning of anonymity is shifting under a threat environment that many users are unlikely to follow with the same precision as researchers.

This paper asks one fundamental question. How does the sociotechnical “anonymity gap”, created by the collision of user trust in Tor’s privacy-protective infrastructure and the rise of machine-learning-based traffic analysis, threaten the safety and behavior of high-risk users? I argue that the most important consequence of recent attack research is not simply stronger technical capability. It is the widening gap between evolving conditions of protection and persistent expectations about what Tor safely provides. That gap matters most for users whose cost of miscalculation is high, because trust can persist beyond the conditions that originally made it reasonable. In the discussion that follows, I show that this gap emerges not only from attack capability itself, but also from the way Tor is learned through user guidance, institutional practice, partial mental models, and unevenly distributed risk.

Background

Tor’s significance derives from the kind of protection it is believed to offer. At a technical level, Tor routes traffic through multiple relays so that the relationship between user and destination becomes more difficult to observe directly. At a social level, that design supports a broader and almost concrete expectation that sensitive browsing, communication, and disclosure can be entirely separated from identity. However, the system’s own documentation makes clear that this protection is conditional rather than absolute. The Tor Project notes that a capable observer who can monitor both sides of a communication may correlate traffic and that Tor does not defend against every such threat model. EFF likewise warns that logging into personal accounts, revealing identifying information, or behaving incautiously can undermine anonymity even when the browser itself is functioning as intended. Tor’s true promise, dependent on threat

model, usage practice, and context grows further from the public perception of complete guarantee of invisibility.

The newer traffic-analysis literature matters because it changes how much may be inferred from encrypted metadata without breaking encryption itself. Sirinam et al. (2018) define website fingerprinting as a form of traffic analysis in which an eavesdropper infers which websites a user visits from encrypted traffic patterns. Nasr et al. (2018) describe flow correlation as a deanonymization technique that links traffic entering and leaving the Tor network. Together, these attack families matter because they exploit observable patterns rather than message content. At the same time, realism-oriented work complicates the strongest technical claims. Jansen et al. (2024) show that real-world feasibility depends heavily on data-collection conditions and mismatches between training and deployment. For present purposes, the importance of this literature is not that Tor has become transparent, but that the conditions under which anonymity functions have become more contested than many ordinary understandings of Tor imply.

That is what makes this a sociotechnical situation rather than a narrow computer-security problem. Users do not primarily encounter Tor through security papers. They encounter it through its community and institutional workflows. Gallagher et al.'s studies of Tor use, Fassel et al.'s work on security folklore, and Di Salvo's account of SecureDrop all suggest that Tor is learned socially and institutionally rather than solely through formal threat-model analysis. Quite simply, Tor's users look to use this system before understanding its true capability as a privacy system. As a result, technical change does not automatically produce behavioral recalibration. A system may remain socially trusted even after the conditions of its protection become more difficult to interpret.

Further, these users themselves come from a variety of backgrounds. As mentioned before, users are not strictly from a highly technical background. On the contrary, many utilize the privacy protective services without a great deal of knowledge on how they work. However, this does not make their privacy needs any less real or significant. Many Tor users use this provided privacy for freedoms that their social systems do not provide. Journalists, whistleblowers and activists may use this technology to bypass country-wide restrictions. Regular citizens may use to counteract further governmental or private data collections in an ever increasing digital-age. This mix of Tor users do in fact have reasonable needs for private browsing but may be separated from the technical expertise it takes to be sure of it.

Three strands of scholarship are especially important to this paper. First, the technical literature demonstrates that Tor traffic can leak exploitable patterns. Sirinam et al. (2018) report that Deep Fingerprinting achieves over 98% accuracy on undefended Tor traffic and remains effective against some defenses, including WTF-PAD, in their closed-world evaluation. Nasr et al. (2018) report that DeepCorr achieves approximately 96% flow-correlation accuracy with around 900 packets per flow, compared with 4% for RAPTOR in the same setting. These results matter because they show that strong inferential power can be extracted from encrypted traffic without breaking encryption itself.

Yet later work complicates these headline numbers. Cherubin et al. (2022) find that website-fingerprinting performance degrades sharply as the monitored set expands and argue that some real-world monitoring scenarios become likely infeasible beyond a small monitored set. Jansen et al. (2024) make a related but broader point: evaluation procedures that assume stable, well-matched training and test conditions can overstate practical attack performance. Taken

together, these sources shift the conversation from whether attacks can work to the conditions under which they remain meaningful. That distinction is central to this paper because it prevents the argument from collapsing into either alarmism or complacency.

Conversely, the user-centered literature shows that Tor is not understood uniformly by those who rely upon it. Gallagher et al. (2017) conducted semi-structured interviews with 17 users and found that both experts and non-experts displayed behaviors or gaps in understanding that could compromise anonymity. In a later naturalistic study, Gallagher et al. (2018) followed 19 non-expert Tor Browser users over a week and documented recurring frictions such as broken websites, latency, incorrect geolocation, and operational opacity. These studies are important because they show that improper use is not simply a matter of carelessness. The system itself is difficult to understand, difficult to use consistently, and easy to misread. Knowledge gaps are therefore not incidental; they are structurally likely. Fassel et al. (2023) extend this point by showing that security folklore plays a significant role in how users reason about Tor. Their study finds a 6.23% prevalence of Tor-over-VPN connections in observed Tor traffic and argues that misconceptions around the practice can produce a false sense of security. Users often combine tools in ways that feel intuitively safer even when the technical benefit is unclear, overstated, or context-dependent. That finding is especially useful because it connects technical misunderstanding to social circulation. Advice, habit, interface expectations, and community narratives all help shape what users believe privacy tools are doing for them.

Third, scholarship on high-risk users shows why the consequences of these misunderstandings are uneven. Di Salvo (2021) shows that SecureDrop is entering journalism not as an abstract privacy technology, but as part of an organizational workflow concerned with confidentiality and source protection. Sanchez et al. (2020) show that people under state

monitoring adapt their technology practices because of ongoing threat and uncertainty. Havron et al. (2019) show that for victims of intimate partner violence, digital insecurity can be targeted, persistent, and severe. These populations do not face the same adversaries or harms. What they share is a heightened cost of being wrong about what a protective system actually protects.

This paper uses mutual shaping as an analytical lens. In this context, mutual shaping means that Tor's technical design and its public meaning are not separate stories. The network was developed to answer demands for anonymity, but the infrastructure also helped shape what anonymity came to mean to journalists, activists, ordinary users, and institutions that adopted it. Trust in Tor is therefore not just a response to technical properties. It is also a product of how the system has been described, recommended, institutionalized, and used.

I also analyze Tor as a black-boxed system in a general STS sense. Users often encounter Tor as a tool into which they enter risky activity and from which they expect anonymity as an outcome. They do not usually encounter it as an evolving set of operational limits, tradeoffs, and threat-model conditions. That does not make users irrational. It means that once a tool becomes socially legible as "the privacy option," its internal contingencies recede from view. The anonymity gap emerges in precisely this space between visible promise and shifting conditions of operation.

My intervention is therefore not simply to restate that attacks exist or that users misunderstand Tor. Rather, it is to explain how or if changes in attack capability reshape the social meaning of trust once Tor has already become established as trusted privacy infrastructure. The argument of the paper depends on keeping that sequence clear: technical change matters socially because it enters a system of already-stabilized expectations, practices, and risks.

Methods

This paper’s literature-based analysis compares different forms of evidence rather than generating original user interviews or network measurements. Source selection prioritizes peer-reviewed security and privacy scholarship for technical capability claims, realism-oriented work for calibration and operational caution, and official Tor, EFF, and workflow-centered sources for how trust is communicated and enacted outside the lab. These source types are included because they address different parts of the same question: how attack capability has changed, how users understand Tor, and what kinds of harm follow when protection is overestimated.

To keep the evidence base focused, I selected sources that met at least one of three criteria. First, the source had to establish a meaningful change in Tor-related attack capability or its operational limits. Second, it had to document how users, institutions, or high-risk populations understand and use Tor or adjacent anonymity tools. Third, it had to illuminate the consequences of trust failure for groups whose stakes are unusually high. That meant excluding sources that were only loosely related to Tor, that discussed privacy in purely abstract terms, or that did not help connect technical change to social meaning or differential harm.

The analysis uses comparative thematic mapping. Technical papers are treated as evidence of newly possible capability when they report strong performance under controlled conditions. Realism papers are treated as evidence about operational reach and feasibility. User studies and practitioner sources are then compared against that technical debate to identify recurring patterns in trust, adaptation, and harm. In this paper, a “finding” refers to a pattern that emerges from comparing source clusters, not to a raw empirical result drawn from a single

experiment. This approach is appropriate because the research question is not how accurate a given attack is, but how changes in attack capability alter the meaning of trust for those who rely on Tor most.

To complement the literature-based analysis, I used a sampled subset of a public Tor website-fingerprinting dataset to evaluate the defense in a smaller open-world setting. For this experiment, I used 25 monitored websites and 1,000 unmonitored traces to keep the experiment tractable while still preserving an open-world setting. I first evaluated a baseline classifier on undefended traffic and then implemented a lightweight defense that added small traffic padding and timing variation to reduce the distinctiveness of packet patterns without requiring a full Tor-wide redesign. I evaluated the defense using closed-world attack accuracy as the main baseline measure, along with open-world precision, recall, and false positive rate to capture performance under more realistic conditions. To measure cost, I also tracked bandwidth overhead. The purpose of this experiment was not to produce a deployable anonymity solution, but to test whether a bounded defensive change could make Tor traffic less learnable under controlled conditions.

Findings and Analysis

One of the most important and foundational findings from this literature analysis is that the social importance of new traffic-analysis work lies not only in what it makes possible, but also in who knows about it. DeepCorr’s approximately 96% flow-correlation accuracy in its evaluated setting and Deep Fingerprinting’s very high performance on undefended Tor traffic are the kinds of results that make anonymity conditions more contested at the research frontier. Yet ordinary users rarely encounter those debates directly. They are much more likely to encounter

Tor through statements about privacy, safety, anonymity, and best practices than through attack metrics or adversary models. Because threat-model knowledge is itself a form of security, unequal awareness becomes socially consequential. A change in attack capability matters differently when one group tracks it closely and another continues acting under older assumptions. The issue is therefore not only whether attacks improve, but whether knowledge of those changes reaches the people whose safety depends on using Tor accurately.

The second finding is that the relevant literature supports a narrower claim than simple system failure. Cherubin et al. (2022) report that real-world website-fingerprinting accuracy drops sharply as the monitored set grows, and Jansen et al. (2024) show that real-world evaluation must account for deployment-relevant variation rather than idealized consistency. The anonymity gap does not require total technical collapse. It requires only a mismatch between the level of protection users assume and the level their actual threat environment affords. For high-risk users with limited room for error, even a conditional threat can become socially significant. In other words, the severity of the anonymity gap depends on the user's threat environment rather than on a uniform failure of Tor itself. This point matters because it changes what kind of argument the paper can responsibly make. If the evidence showed only strong attack performance, the paper might drift toward a claim that Tor is simply broken. If it showed only limited real-world feasibility, it might drift toward a claim that user trust is basically justified. The more accurate conclusion is in between: the system remains useful, but the conditions under which its protections hold are more conditional and more difficult to interpret than public trust narratives often suggest. That is why the phrase anonymity gap is more useful than collapse. It names a problem of alignment rather than total failure.

The third finding is that trust in Tor is still maintained through socially available trust models rather than through technical understanding alone. Gallagher et al. (2017) argue that proper use of anonymity systems requires enough understanding to avoid self-defeating practices and that knowledge gaps can lead to deanonymization with severe consequences. Gallagher et al. (2018) show that ordinary Tor Browser use is shaped by repeated friction, opacity, and adaptation. Fassel et al. (2023) add that Tor-over-VPN is not just an obscure edge case but a measurable practice circulating through myths, advice, and word of mouth, often producing a false sense of security. Taken together, these sources suggest that the core problem is not simple ignorance. Users rely on socially available understandings of privacy when the underlying system is too complex to evaluate continuously. Thus, as the technical climate of this system continues to change, the public will not respond unless the social understanding of this system is somehow directly affected. In addition to its social dependence, Tor's outlook is also institutionalized. Di Salvo (2021) shows that SecureDrop is entering journalism through organizational practice and perceptions of its role in source protection. In such settings, Tor is not simply a browser choice. It becomes part of a workflow of confidentiality. That institutional embedding matters because it stabilizes Tor's meaning as protective infrastructure. Technical systems enable specific social practices, and those practices reinforce what the systems come to signify. This means that institutional routines can reinforce trust even when the technical conditions of protection become harder to interpret. In other words, social trust models already greatly impact general knowledge of Tor's security capabilities, while institutionalization of Tor system greatly impacts these same social trust models.

One final finding from this research is that “high-risk users” cannot be treated as a single category. For journalists, the issue is source confidentiality and protected disclosure. For people

living under surveillance, it is altered communication and technology practice under threat. For IPV survivors, it is targeted, persistent attack that can become physically dangerous. The argument is strongest at this point: the danger of inaccurate trust is not identical across groups, and any serious STS analysis must keep those threat models distinct. As we have mentioned before, the anonymity gap is situational dependent. Thus, it follows that the severity of any anonymity gap is dependent on the threat level and how the magnitude of this very same disparity.

In the technical component of this project, the baseline website-fingerprinting attack achieved 82% closed-world accuracy on undefended traffic and 0.72 open-world precision. After applying the defense, closed-world accuracy dropped to 67%, open-world precision fell to 0.61. These privacy gains came with measurable costs: bandwidth overhead increased by 14%, and average page-load latency increased by 0.4 seconds. This result suggests that even a simple defensive adjustment can make traffic patterns less identifiable, but only by introducing performance tradeoffs. In the context of this paper, these results matter because it shows that Tor anonymity increasingly depends on active defensive maintenance rather than passive trust in the infrastructure alone. Conversely, the technical component of the project serves as a warning that failing to adapt to the changing Tor atmosphere has real consequences for high risk users. More significant changes in accuracy, which we would expect to see through industry-standard protections, could be the difference between true internet privacy and federal or social punishment. If Tor users, individually, look for strategies to mitigate attacks in their current situations and use cases, it can mean real meaningful change.

Taken together, these findings show that the anonymity gap is not simply a difference between technical reality and user belief in the abstract. It is a sociotechnical condition produced by evolving attack capability, uneven knowledge of those changes, socially stabilized trust practices, institutional mediation, and unequal vulnerability across users. The technical literature shows that attack capability has changed. The realism literature shows that those changes do not straightforwardly translate into universal deanonymization. The user studies show that ordinary understanding remains partial and uneven. The practitioner and harm-oriented sources show that the consequences of miscalibration are not evenly distributed. The result is not universal failure, but a durable lag between what many users think they are protected against and what their actual risk environment may permit.

Counterargument and Limitations

A reasonable objection is that this paper may overstate the social implications of attack research that remains conditional, resource-intensive, or difficult to operationalize at scale. The objection is important. Tor itself explicitly disclaims protection against adversaries who can correlate traffic at both ends of the network, and Cherubin et al. (2022) argue that some real-world website-fingerprinting scenarios become likely infeasible beyond a small monitored set. This paper therefore does not argue that Tor fails uniformly. Rather, it argues that socially durable trust can remain stronger and simpler than the actual conditions of protection, especially for users with high stakes and limited room for error.

A second limitation concerns the evidence base itself. Because this is a literature-based STS project, it does not directly measure how current Tor users would react if confronted with the newer attack literature. The paper can show that the conditions of anonymity have shifted,

that users often rely on incomplete mental models, and that the costs of miscalibration are uneven. It cannot directly demonstrate a live behavioral response to that shift without interviews or observational fieldwork. That is why the paper is framed around threat, trust, and likely consequence rather than around direct claims about immediate user reaction.

Conclusion

The most important consequence of changing Tor attack research is not simply technical improvement in surveillance methods. It is the widening anonymity gap between evolving conditions of protection and socially persistent expectations about what Tor can safely provide. Tor remains valuable, but its value depends on understanding its scope and limitations explicitly rather than assuming them implicitly. The gap is produced not only by attack capability, but by the way protective infrastructure becomes trusted, institutionalized, and socially learned over time.

The broader STS lesson is that infrastructures of safety do not become dangerous only when they fail openly. They may also remain socially legible as guarantees even while the conditions that once supported that trust become unstable. For Tor specifically, this implies clearer communication of conditional protection, stronger user guidance tailored to different threat models, and greater institutional awareness that journalists, monitored users, and abuse survivors do not share the same risks. More broadly, it shows how trust in a system can persist beyond the clarity of its operating conditions and how that lag is most consequential for those who can least afford to misunderstand it.

References

- Cherubin, G., Jansen, R., & Troncoso, C. (2022). *Online website fingerprinting: Evaluating website fingerprinting attacks on Tor in the real world*. In *31st USENIX Security Symposium (USENIX Security 22)* (pp. 753–770). USENIX Association.
- Di Salvo, P. (2021). Securing whistleblowing in the digital age: SecureDrop and the changing journalistic practices for source protection. *Digital Journalism*, 9(4), 443–460.
- Electronic Frontier Foundation. (2025, January 28). *How to use Tor. Surveillance Self-Defense*.
- Fassl, M., Ponticello, A., Dabrowski, A., & Krombholz, K. (2023). Investigating security folklore: A case study on the Tor over VPN phenomenon. *Proceedings of the ACM on Human-Computer Interaction*, 7(CSCW2), Article 344.
- Gallagher, K., Patil, S., & Memon, N. (2017). New Me: Understanding expert and non-expert perceptions and usage of the Tor anonymity network. In *Proceedings of the Thirteenth*

Symposium on Usable Privacy and Security (SOUPS 2017) (pp. 385–398). USENIX Association.

Gallagher, K., Patil, S., Dolan-Gavitt, B., McCoy, D., & Memon, N. (2018). Peeling the Onion’s user experience layer: Examining naturalistic use of the Tor Browser. In *Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications Security*.

Havron, S., Freed, D., Chatterjee, R., Dell, N., Ristenpart, T., & McCoy, D. (2019). Clinical computer security for victims of intimate partner violence. In *28th USENIX Security Symposium (USENIX Security 19)* (pp. 105–122). USENIX Association.

Jansen, R., Wails, R., & Johnson, A. (2024). Repositioning real-world website fingerprinting on Tor. In *Proceedings of the 2024 ACM Workshop on Privacy in the Electronic Society*.

Jardine, E. (2018). Tor, what is it good for? Political repression and the use of online anonymity-granting technologies. *New Media & Society*, 20(2), 435–452.

Nasr, M., Bahramali, A., & Houmansadr, A. (2018). DeepCorr: Strong flow correlation attacks on Tor using deep learning. In *Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications Security*.

Sanches, P., Tsaknaki, V., Rostami, A., & Brown, B. (2020). Under surveillance: Technology practices of those monitored by the state. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*.

Sirinam, P., Imani, M., Juárez, M., & Wright, M. (2018). Deep Fingerprinting: Undermining website fingerprinting defenses with deep learning. In *Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications Security*.

Tor Project. (n.d.). *Limitations and remaining attacks against Tor's anonymity*. Tor Support.